

Testing of Hypothesis: Several Proportions

Reading Objectives:

In this reading, you will learn about testing hypotheses with several proportions. This involves nonparametric related hypotheses. The chi-square test and F-test are statistics used in these validations.

Main Reading Section:

Testing of Hypothesis: Several Proportions

Suppose the hypothesis involves several populations and corresponding samples. Based on the sample proportion, we will try to validate the claim or hypothesis.

We use the chi-square test as the test statistic given by

$$\chi^2 = \sum \frac{(\mathbf{O} - \mathbf{E})^2}{\mathbf{E}}$$

Where O is observed frequencies, and E is expected frequencies.

Degrees of freedom = $(r - 1) \times (c - 1)$, where r is the number of rows and c is the number of columns.